Basic Electrical Engineering – Complete Notes (B.Tech CSE – AI, Semester 1, BBDU)

Unit 1: Basics of Electrical Engineering

- Definition of electrical engineering
- Charge, Current, Voltage, Power & Energy
- Ohm’s Law & its limitations
- Types of elements: Active & Passive
- KCL & KVL (Kirchhoff’s Current & Voltage Laws)
- Series & Parallel circuits
- Star & Delta connections basics

Unit 2: Network Theorems

- Superposition Theorem
- Thevenin's Theorem
- Norton's Theorem
- Maximum Power Transfer Theorem
- Source Transformation

Unit 3: AC Fundamentals

- AC vs DC, sinusoidal waveforms
- Instantaneous, average, RMS values
- Form factor & peak factor
- Phasor representation
- Power in AC: Active, Reactive & Apparent
- Power factor & its importance

Unit 4: Electrical Machines

- Basic magnetic circuit concepts
- Single-phase transformer: working principle & EMF equation
- Losses in transformer, efficiency
- DC generators & motors: working principle, types

Unit 5: Measuring Instruments & Power Systems

- Ammeter, Voltmeter, Wattmeter, Energy meter
- Basics of power generation (Thermal, Hydro, Solar, Nuclear)
- Transmission & distribution basics
- Electrical safety precautions

These notes follow the structure used in most Indian engineering syllabi including BBDU for Semester■1 Basic Electrical Engineering.